

UNIT 6: DIVERSITY OF LIFE FORMS

6.7 – VERTEBRATES

Vertebrate Outline

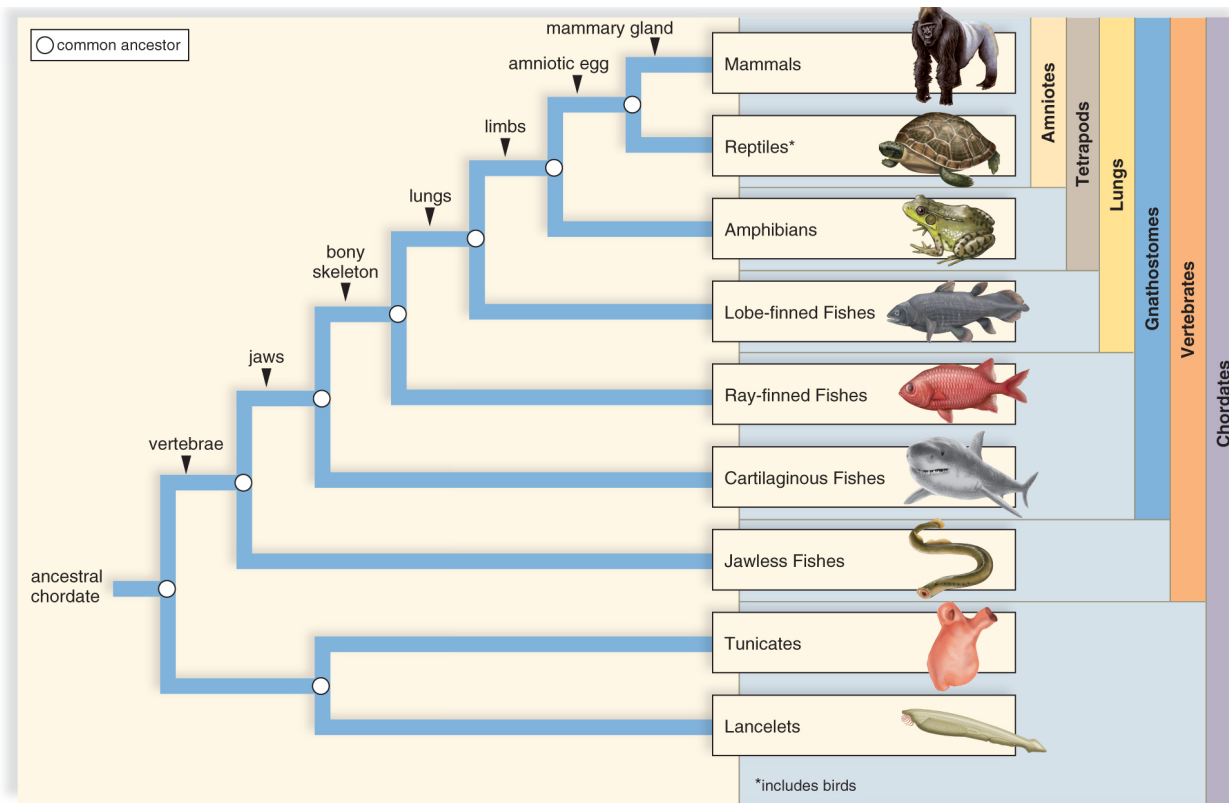
- Chordates
 - Nonvertebrate vs Vertebrate Chordates
- Characteristics of Vertebrates
- Vertebrate diversity
 - Fish
 - Amphibians
 - Reptiles & Birds
 - Mammals

Characteristics of Chordates

- Deuterostomes, with internal bone and cartilage skeleton instead of exoskeleton
- All chordates have 4 basic characteristics:
 - **Notochord** – dorsal supporting rod, turns into vertebral column in many
 - **Dorsal tubular nerve cord** – Anterior portion becomes the brain. Known as spinal cord in many
 - **Pharyngeal pouches** – Observable in embryos for most. Become gills in fish, varying function in others
 - **Postanal tail** – Tail structure present at some point

Nonvertebrate Chordates

- Do not have a spine made of bony vertebrae
- Lancelets (cephalochordates)
 - ▣ Marine Chordates
 - ▣ Maintain all 4 characteristics of chordates for entire life
- Sea squirts (urochordates)
 - ▣ Also known as tunicates
 - Have an outer, tunic-like covering
 - Larva possesses all 4 characteristics
- You can see the phylogenetic tree of all chordates here

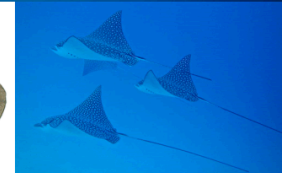
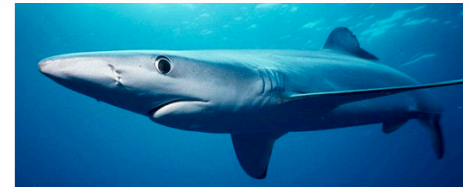


Vertebrate Characteristics

- Have same 4 characteristics as non-vertebrates as embryos, but also all share other characteristics
 - ▣ **Vertebral column** – replaces the notochord
 - ▣ **Skull** – contains brain and houses other sensory features
 - ▣ **Endoskeleton** – living tissue that grows with the organism, protects organs, anchor for muscles
 - ▣ **Internal organs** – large coelom and complete digestive tract, closed circulatory system, respiration using lungs or gills, sexes generally separate, with sexual reproduction
- Evolution
 - ▣ Most likely began with jawless fish
 - ▣ Fish then developed jaws
 - ▣ Amphibians most likely first terrestrial organisms with limbs, and reproduced and lived partial lives in aquatic environments
 - ▣ Reptiles first fully adapted to land, and first organisms to have amniotic eggs. Also have watertight skin
 - ▣ Birds and mammals evolved further from there

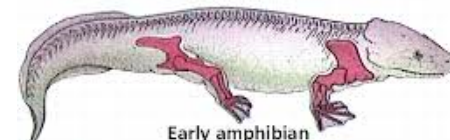
Diversity – Fish

- Largest group of vertebrates. mm in length to 12m in length
 - ▣ Jawless Fish (Agnathans)
 - Cylindrical, up to a meter long, smooth, nonscaly skin.
 - Scavengers that feed off dead fish
 - Lamprey
 - ▣ Fish with jaws
 - All share following characteristics
 - Ectotherms (environment regulates temp), gills, bony endoskeleton, and scales
 - Extinct placoderms
 - ▣ Cartilaginous fish
 - Cartilage skeleton instead of bone
 - 5-7 gill slits, no gill cover
 - Teethlike scales
 - Sharks, rays, skates, chimaeras



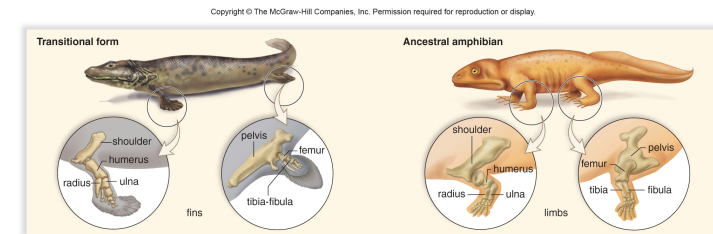
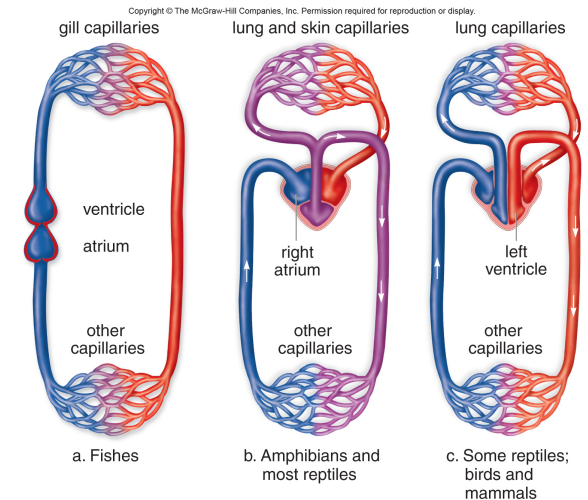
Diversity – Fish

- Largest group of vertebrates.
mm in length to 12m in length
 - Bony Fish
 - Most are ray finned (fan shaped)
 - Have swim bladder for buoyancy
 - Well developed nervous system and brain
 - Lobe-finned fish
 - Contain fleshy fins supported by bone
 - Gave rise to first amphibians
 - Contain lungfish and coelacanth



Diversity – Amphibians

- Live life on both land and water, with the following unifying characteristics:
 - ▣ **Limbs** – normally tetrapods (4 limbs)
 - ▣ **Smooth, non-scaly skin** – for maintaining water balance
 - ▣ **Lungs** – if present, small and used for gas exchange
 - ▣ **Double-looped circulatory pathway** – 3 chamber heart
 - ▣ **Sense organs** – sight, hearing, and smell tuned for life on land
 - ▣ **Ectotherms**
 - ▣ **Aquatic reproduction** – Deposit jelly-coated eggs in water and are fertilized; young hatch and grow early part of life in water. Most emerge for life on land as adults
- Diversity includes
 - ▣ Salamanders and newts
 - ▣ Frogs and Toads
 - ▣ Caecilians

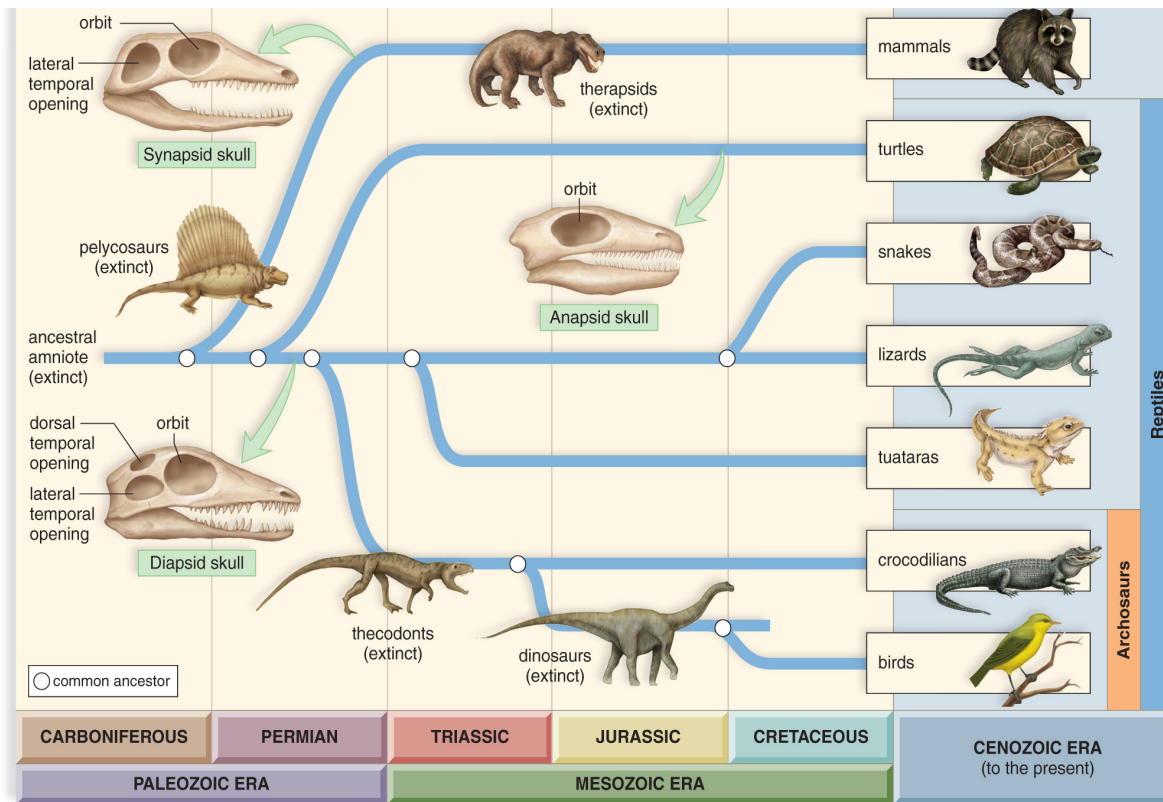


Diversity - Reptiles

- Terrestrial animals, which includes birds in class Reptilia, with the following unifying characteristics:
 - ▣ **Paired limbs** – 2 pairs, usually 5 toes each. Adapted for climbing, running, paddling, or flying
 - ▣ **Skin** – Thick and dry, impermeable to water, scaly
 - ▣ **Efficient breathing** - lungs better developed than amphibians
 - ▣ **Efficient circulation** – Heart prevents mixing of blood
 - ▣ **Efficient excretion** – well developed kidneys
 - ▣ **Ectotherms**
 - ▣ **Well-adapted reproduction** – internal fertilization. Amniotic egg
- Amphibians gave rise to reptiles according to multiple lineages

Evolution of Reptiles

- Lineages grouped on presence and number of openings behind eye-socket
 - Anapsid – no opening
 - Turtles
 - Diapsid – 2 openings
 - Most reptiles and birds
 - Synapsid – 1 opening
 - Mammals
- Most reptiles do not have 1 known common ancestor – paraphyletic rather than monophyletic
- Crocodiles and birds more closely related to dinosaurs than lizards



Diversity of Reptiles and Birds

□ Living reptiles include:

- ▣ Tortoises and turtles
- ▣ Lizards
- ▣ Snakes
- ▣ Tuataras
- ▣ Crocodilians

□ Birds are also included and have features such as:

- ▣ **Feathers**
- ▣ **Modified skeleton**
- ▣ **Modified respiration (with lobular lungs)**
- ▣ **Endotherms** –metabolic heat to maintain temp
- ▣ **Well-dev. sense organs and nervous system**
- ▣ **Diversity includes:**
 - Flying birds – eagles, geese, mockingbirds
 - Flightless - emu, kiwi, penguin, ostrich
 - Normally classified by beak and foot type, and habitat



Diversity - Mammals

- Include largest animals to live, with the following unifying characteristics:
 - ▣ **Hair** – ornamental or purpose
 - ▣ **Mammary glands**
 - ▣ **Skeleton** – holds larger brain than reptiles
 - ▣ **Internal Organs** – efficient resp and circ, 4 chamber heart, developed nervous system
 - ▣ **Internal development** – born alive after development
- Mammalian evolution based on type of offspring birth state
 - ▣ Monotremes – egg laying mammal
 - Only includes duck-billed platypus & Echidna
 - ▣ Marsupials – Pouched mammals (embryos nourished by a yolk based placenta in pouch)
 - Kangaroo, koala, wombats, opossums
 - ▣ Placental – Placenta internal for full growth and development
 - All other animals



Diversity of Mammals

For more info on the full diversity of mammals, see page 566 in text

Order	Typical Examples		Key Characteristics	Approximate Number of Living Species
Rodentia	Beavers, mice, porcupines, rats		<i>Small plant eaters</i> Chisel-like incisor teeth	1,814
Chiroptera	Bats		<i>Flying mammals</i> Primarily fruit or insect eaters; elongated fingers; thin wing membrane; nocturnal; navigate by sonar	986
Insectivora	Moles, shrews		<i>Small, burrowing mammals</i> Insect eaters; most primitive placental mammals; spend most of their time underground	390
Marsupialia	Kangaroos, koalas		<i>Pouched mammals</i> Young develop in abdominal pouch	280
Carnivora	Bears, cats, raccoons, weasels, dogs		<i>Carnivorous predators</i> Teeth adapted for shearing flesh; no native families in Australia	240
Primates	Apes, humans, lemurs, monkeys		<i>Tree-dwellers</i> Large brain size; binocular vision; opposable thumb; end product of a line that branched off early from other mammals	233
Artiodactyla	Cattle, deer, giraffes, pigs		<i>Hoofed mammals</i> With two or four toes; mostly herbivores	211
Cetacea	Dolphins, porpoises, whales		<i>Fully marine mammals</i> Streamlined bodies; front limbs modified into flippers; no hind limbs; blowholes on top of head; no hair except on muzzle	79
Lagomorpha	Rabbits, hares, pikas		<i>Rodent-like jumpers</i> Four upper incisors (rather than the two seen in rodents); hindlegs often longer than forelegs, an adaptation for jumping	69
Suborder Pinnipedia	Sea lions, seals, walrus		<i>Marine carnivores</i> Feed mainly on fish; limbs modified for swimming	34
Edentata	Anteaters, armadillos, sloths		<i>Toothless insect eaters</i> Many are toothless, but some have degenerate, peglike teeth	30
Perissodactyla	Horses, rhinoceroses, zebras		<i>Hoofed mammals with one or three toes</i> Herbivorous teeth adapted for chewing	17
Proboscidea	Elephants		<i>Long-trunked herbivores</i> Two upper incisors elongated as tusks; largest living land animal	2